



Practice Questions

Simplification

Directions (Q.1-5): What value should come in place of the question-mark(?) in the following question

1. $45\% \text{ of } 200 + \sqrt{?} = 30\% \text{ of } 700 - 18\% \text{ of } 300$

- 1) 4250
- 2) 4356
- 3) 4365
- 4) 4300
- 5) none of these

2. $\sqrt{1936} + \sqrt{3025} = ? \% \text{ of } 220$

- 1) 41
- 2) 45
- 3) 50
- 4) 52
- 5) none of these

3. $\frac{3}{8} \text{ Of } 720 \div 15 + 12 = \sqrt{?}$

- 1) 904
- 2) 900
- 3) 905
- 4) 890
- 5) none of these

4. $\frac{7}{15} \text{ of } \frac{5}{11} \text{ of } \frac{8}{17} \text{ of } \frac{5}{19} \text{ of } 639540 = ?$

- 1) 16800
- 2) 16550
- 3) 160800
- 4) 16500
- 5) none of these

5. $6269 + 0.25 \times 444 + 0.8 \times 200 = ? \times 15$

- 1) 436
- 2) 426
- 3) 446
- 4) 450
- 5) none of these

Directions (Q.6-10): What value should come in place of the question-mark(?) in the following question:

6. $\sqrt{10201} \times \sqrt{3969} - (52)^2 = \sqrt{?} + (60)^2$

- 1) 3481
- 2) 3520
- 3) 3490
- 4) 3580
- 5) None of these

7. $0.7 \times 7.7 \times 7.77 = ?$

- 1) 41.8033
- 2) 41.8080
- 3) 41.8883
- 4) 41.8803
- 5) None of these

8. $16\frac{2}{3} \times \frac{5}{7} + 6\frac{2}{3} \times \frac{11}{14} = ?$

- 1) $17\frac{1}{7}$
- 2) $17\frac{3}{5}$
- 3) $17\frac{2}{7}$
- 4) $17\frac{5}{7}$
- 5) None of these

9. $\frac{6}{17}$ of $\frac{5}{13}$ of $\frac{4}{11}$ of $\frac{7}{11}$ of 427856 = ?

- 1) 13400
- 2) 13440
- 3) 13200
- 4) 13000
- 5) None of these

10. $\frac{5}{18}$ Of $8802 \div 5 + 51 = \sqrt{?}$

- 1) 2921600
- 2) 2921500
- 3) 2921670
- 4) 2921660
- 5) None of these

Direction (Q.11-15): What should come in place of the question mark (?) in the following questions?

11. $(1296)^{-\frac{3}{4}} = ?$

- 1) $\frac{1}{6^2}$
- 2) $\frac{1}{6^2}$
- 3) $\frac{1}{6^5}$
- 4) $\frac{1}{6^3}$
- 5) None of these

12. $(5^4)^5 \times (25^3)^3 = ?$

- 1) 5^{38}
- 2) 5^{22}
- 3) 5^{49}
- 4) 5^{52}
- 5) None of these

13. $\left(\frac{256}{6561}\right)^{\frac{1}{4}} = ?$

- 1) $\frac{4}{9}$
- 2) $\frac{5}{7}$
- 3) $\frac{4}{3}$
- 4) $\frac{2}{9}$
- 5) None of these

14. $(.015625)^{\frac{5}{6}} = ?$

- 1) $\frac{2^5}{10^5}$
- 2) $\frac{2^5}{18^5}$
- 3) $\frac{5^5}{11^4}$
- 4) $\frac{5^5}{10^5}$
- 5) None of these

15. $7^2 \times 6^0 \times 2^2 \times 3^2 = ?$

- 1) 1672
- 2) 1780
- 3) 1764
- 4) 1690
- 5) None of these

Direction (Q16-20): What will come in the place of question mark (?) in the following given questions:

16. $672 \div 28 \times 24 + 363 - 309 = ?$

- 1) 360
- 2) 600
- 3) 630
- 4) 300
- 5) None of these

17. $\sqrt{19.36} + \sqrt{14.44} + \sqrt{37.21} = ? \times 1.3$

- 1) 10
- 2) 13
- 3) 21
- 4) 17
- 5) 11

18. $74\% \text{ of } 2840 + 80\% \text{ of } 1640 - ? = 47^2$

- 1) 1042.6
- 2) 1240.6
- 3) 1420.6
- 4) 1204.6
- 5) None of these

19. $(2.5)^8 \times (6.25)^4 \div (15.625)^3 = 2.5^?$

- 1) 1
- 2) 4
- 3) 8
- 4) 6
- 5) 7

20. $4.004 + 5.7(2.5 - 0.5) = ?$

- 1) 25.404
- 2) 15.404
- 3) 51.004
- 4) 15.004
- 5) 11.004

Directions(Q21-25): What should come in place of the question mark (?) in the following questions?

21. $22\% \text{ of } 280 + 34\% \text{ of } 1080 \times \frac{5}{12} = ? + 16\% \text{ of } 460$

- 1) 141
- 2) 181
- 3) 152
- 4) 183
- 5) 158

22. $\left[\sqrt[3]{\frac{3375}{19683}} - \sqrt[3]{\frac{125}{1728}} \right] \div \sqrt[3]{\frac{64}{729}} = ? - \frac{3}{8}$

- 1) $\frac{5}{16}$
- 2) $\frac{3}{16}$
- 3) $\frac{7}{8}$
- 4) $\frac{11}{16}$
- 5) None of these

23. $(22^2 \times 8^2) \div (92.4 \div 4.2) = ? \times 32$

- 1) 22
- 2) 44
- 3) 36
- 4) 24
- 5) 32

24. $(144 \div 4)^2 \times (72 \div 12)^3 = 12 \times ? \times (25920 \div 10)$

- 1) 6
- 2) 12
- 3) 9
- 4) 16
- 5) 24

25. $808 \div (128)^{\frac{1}{7}} + 482 = 4 \times ? + 846$

- 1) 40
- 2) 10
- 3) 20
- 4) 8
- 5) 5

Directions (Q.26-30): What should come in place of question mark (?) in the following number question?

26. $\frac{15^2 \times 24 + 2540}{25} = 44464 \div ?$

1) 120

2) 135

3) 130

4) 140

5) 145

27. $\frac{7}{11} \times 1034 + 1\frac{4}{7} \times 2401 = 1230 + ?$

1) 3201

2) 3102

3) 5021

4) 4021

5) 3210

28. $[(25.5 \times 38) \div 57 - 1640 \div 410] = ? \div \times 13$

1) 0.2

2) 1

3) 2

4) 4

5) 1.5

29. $(3333 \div 30) + (785 \div 25) + (2981 \div 11) = ?$

- 1) 413.5
- 2) 523.5
- 3) 415
- 4) 513.5
- 5) None of these

30. $43.5 \times \left[\frac{1}{7} \times (91 - 49) + 17\frac{1}{3} \right] = ?$

- 1) 1051
- 2) 1015
- 3) 2015
- 4) 2501
- 5) 1501

Directions (Q 31-35): What value should come in place of the question-mark(?) in the following questions:

31. $52\% \text{ of } 400 + \sqrt{?} = 60\% \text{ of } 600 - 25\% \text{ of } 400$

- 1) 2704
- 2) 2710
- 3) 2712
- 4) 2700
- 5) None of these

32. $\sqrt{10000} \times \sqrt{8100} - (50)^2 = \sqrt{?} + (80)^2$

- 1) 19600
- 2) 16900
- 3) 14400
- 4) 10000
- 5) None of these

33. $\frac{2}{5}$ of $\frac{3}{4}$ of $\frac{7}{9}$ of 7200 = ?

- 1) 1680
- 2) 1608
- 3) 1688
- 4) 1670
- 5) None of these

34. $135 \div 15 \times 19 + 14807 = ? + \sqrt{3249} - \sqrt{9604}$

- 1) 15019
- 2) 16157
- 3) 13369
- 4) 17843
- 5) None of these

35. $\frac{3}{7}$ Of $504 \div 12 + 17 = \sqrt{?}$

- 1) 35
- 2) 1225
- 3) 625
- 4) 125
- 5) none of these

Direction(Q36- 40): What will come in place of question mark?

36. $648 \div 36 \times 49 - 1012 + 847 = ?$

- 1) 825
- 2) 717
- 3) 669
- 4) 707
- 5) None of these

37. $2048 \times 1824 \div 76 = (? - 212) \times 64$

- 1) 855
- 2) 1100
- 3) 980
- 4) 778
- 5) None of these

38. $\sqrt{24^2 + 285 - 8^2 - 172} = ?^2$

- 1) 64
- 2) 25
- 3) 8
- 4) 5
- 5) None of these

39. $\frac{1}{(32768)^{-\frac{2}{5}}} + \frac{1}{(1296)^{-\frac{3}{4}}} + \frac{1}{(6561)^{-\frac{1}{8}}} = ?$

- 1) 283
- 2) 199
- 3) 323
- 4) 223
- 5) None of these

40. 36% of 640 – 12.5% of 352 + 25% of 640 = ? – 48% of 432

- 1) 558
- 2) 553.76
- 3) 628.6
- 4) 650
- 5) None of these

Directions (Q.41-45): What will come in the place of question mark (?) in the following given question:

41. 84% of 8400 + 42% of 6120 = ?

- 1) 9662.4
- 2) 9266.4
- 3) 6926.4
- 4) 9626.4
- 5) None of these

42. $2\frac{1}{3} + 4\frac{1}{4} + 4\frac{2}{3} + 8\frac{7}{6} + ? = 4\frac{3}{5} \times 4\frac{1}{2}$

- 1) $\frac{17}{60}$
- 2) $\frac{13}{60}$
- 3) $\frac{27}{20}$
- 4) $\frac{17}{50}$
- 5) None of these

43. 36.76 + 2894.713 + 34965.11 = ?

- 1) 38796.583
- 2) 37968.853
- 3) 38796.538
- 4) 37886.385
- 5) 37896.583

44. $\sqrt{4761} \div 23 + \sqrt{12769} = ? \times 58$

- 1) 1
- 2) 0.5
- 3) 1.5
- 4) 2
- 5) 2.5

45. $(512)^{1.3} \times (64)^{1.8} \div (8)^{0.5} = (8)^?$

- 1) 3
- 2) 5
- 3) 4
- 4) 7
- 5) None of these

Directions (Q.46-50): What will come in the place of question mark (?) in the following given question:

46. $32 \times 3 (54 - 15) + 186 \div 3 \div 2 - (21)^2 = ?$

- 1) 3334
- 2) 3434
- 3) 3433
- 4) 3324
- 5) 3343

47. $2850 \div 2.5 - ? \times 42 = 300$

- 1) 10
- 2) 30
- 3) 40
- 4) 20
- 5) None of these

48. $\frac{21 \times 51 + 54}{9 \times 14 - 30} = ?$

1) $21\frac{13}{23}$

2) $11\frac{13}{22}$

3) $12\frac{23}{32}$

4) $11\frac{23}{32}$

5) $21\frac{13}{12}$

49. 95% of 830 - ? % of 2770 = 650

1) 3

2) 5

3) 1

4) 2

5) 6

50. $255 \times 8 + 386 \times 5 = ?$ % of 7940

1) 30

2) 50

3) 40

4) 10

5) 20

Solutions

Sol- 1. 2

Sol- 2. 2

Sol- 3. 2

Sol- 4. 1

Sol- 5. 1

Sol- 6. 1

Sol- 7. 4

Sol- 8. 1

Sol- 9. 2

Sol- 10. 5; 291600

Sol- 11. 4:

$$\left(\frac{1}{6^4}\right)^{\frac{3}{4}} = \frac{1}{6^3}$$

Sol- 12. 1;

$$5^{20} \times 5^{18} = 5^{38}$$

Simplification

Sol- 13. 1;

$$\frac{(4)^{\frac{4 \times 1}{4}}}{(9)^{\frac{4 \times 1}{4}}} = \frac{4}{9}$$

Sol- 14. 4

$$\left(\frac{15625}{1000000}\right)^{\frac{5}{6}} = \frac{5^6 \times \frac{5}{6}}{10^6 \times \frac{5}{6}} = \frac{5^5}{10^5}$$

Sol- 15. 3;

Sol- 16. 3

Sol- 17. 5

Sol- 18. 4

Sol- 19. 5

Sol- 20. 2

Sol- 21. 1

$$\frac{22}{100} \times 280 + \frac{34}{100} \times 1080 \times \frac{5}{12} = ? + \frac{16}{100} \times 460$$

$$61.6 + 153 = ? + 73.6$$

$$? = 153 - 73.6 + 61.6$$

$$? = 153 - 12$$

$$? = 141$$

Simplification

Sol- 22. 4

$$\left(\frac{15}{27} - \frac{5}{12}\right) \times \frac{9}{4} = ? - \frac{3}{8}$$

$$\left(\frac{60-45}{108}\right) \times \frac{9}{4} = ? - \frac{3}{8}$$

$$\frac{15}{108} \times \frac{9}{4} = ? - \frac{3}{8}$$

$$\frac{5}{16} = ? - \frac{3}{8}$$

$$? = \frac{5}{16} + \frac{3}{8}$$

$$? = \frac{11}{16}$$

Sol- 23. 2

$$(484 \times 64) \div 22 = ? \times 32$$

$$\frac{484 \times 64}{22} = ? \times 32$$

$$? = \frac{484 \times 64}{22 \times 32}$$

$$? = 44$$

Sol- 24. 3

$$\left(\frac{144}{4}\right)^2 \times \left(\frac{72}{12}\right)^3 = 12 \times ? \times 2592$$

$$36^2 \times 6^3 = 12 \times ? \times 2592$$

$$? = \frac{36 \times 36 \times 6 \times 6 \times 6}{12 \times 2592}$$

$$? = 9$$

Simplification

Sol- 25. 2

$$808 \div (2^7)^{\frac{1}{7}} + 482 = 4 \times ? + 846$$

$$\frac{808}{2} + 482 - 846 = 4 \times ?$$

$$404 + 482 - 846 = 4 \times ?$$

$$886 - 846 = 4 \times ?$$

$$? = \frac{40}{4}$$

$$? = 10$$

Sol- 26. 4

$$\frac{225 \times 24 + 2540}{25} = 44464 \div ?$$

$$\frac{5400 + 2540}{25} = 44464 \div ?$$

$$\frac{7940}{25} = 44464 \div ?$$

$$317.6 = 44464 \div ?$$

$$? = \frac{44464}{317.6}$$

$$? = 140$$

Simplification

Sol- 27. 1

$$\frac{7}{11} \times 1034 + 1\frac{4}{7} \times 2401 = 1230 + ?$$

$$7 \times 94 + 11 \times 343 = 1230 + ?$$

$$658 + 3773 = 1230 + ?$$

$$? = 4431 - 1230$$

$$? = 3201$$

Sol- 28. 2

$$[(25.5 \times 38) \div 57 - 1640 \div 410] = ? \div \times 13$$

$$\left[\frac{969}{57} \right] - \frac{1640}{410} = ? \times 13$$

$$17 - 4 = ? \times 13$$

$$13 = ? \times 13$$

$$? = 1$$

Sol- 29. 1

$$(3333 \div 30) + (785 \div 25) + (2981 \div 11) = ?$$

$$\frac{3333}{30} + \frac{785}{25} + \frac{2981}{11} = ?$$

$$111.1 + 31.4 + 271 = ?$$

$$? = 413.5$$

Simplification

Sol- 30. 2

$$43.5 \times \left[\frac{1}{7} \times (91 - 49) + 17\frac{1}{3} \right] = ?$$

$$43.5 \times \left[\frac{1}{7} \times 42 + \frac{52}{3} \right] = ?$$

$$43.5 \times \left[6 + \frac{52}{3} \right] = ?$$

$$? = 43.5 \times \frac{70}{3}$$

$$? = 14.5 \times 70$$

$$? = 1015$$

Sol- 31. 1

Sol- 32. 4

Sol- 33. 1

Sol- 34. 1

Sol- 35. 2

Sol- 36. 2

$$648 \div 36 \times 49 - 1012 + 847$$

$$\Rightarrow 18 \times 49 - 1012 + 847$$

$$\Rightarrow 1882 - 1012 + 847$$

$$\Rightarrow 717$$

Simplification

Sol- 37. 3

$$2048 \times 1824 \div 76 = (? - 212) \times 64$$

$$\Rightarrow 2048 \times 24 = (? - 212) \times 64$$

$$\Rightarrow \frac{2048 \times 24}{64} = ? - 212$$

$$\Rightarrow 768 = ? - 212$$

$$\Rightarrow ? = 768 + 212 = 980$$

Sol- 38. 4

$$\sqrt{24^2 + 285 - 8^2 - 172} = ?^2$$

$$\Rightarrow \sqrt{576 + 285 - 64 - 172} = ?^2$$

$$\Rightarrow \sqrt{625} = ?^2$$

$$\Rightarrow 25 = ?^2$$

$$? = 5$$

Sol- 39. 1

$$\frac{1}{(32768)^{-\frac{2}{5}}} + \frac{1}{(1296)^{-\frac{3}{4}}} + \frac{1}{(6561)^{-\frac{1}{8}}}$$

$$\Rightarrow \frac{1}{(8^5)^{-\frac{2}{5}}} + \frac{1}{(6^4)^{-\frac{3}{4}}} + \frac{1}{(3^8)^{-\frac{1}{8}}}$$

$$\Rightarrow \frac{1}{(8)^{-5 \times \frac{2}{5}}} + \frac{1}{(4)^{-4 \times \frac{3}{4}}} + \frac{1}{(3)^{-8 \times \frac{1}{8}}}$$

$$\Rightarrow 8^2 + 6^3 + 3$$

$$\Rightarrow 64 + 216 + 3 = 283$$

Simplification

Sol- 40. 2

$$36\% \text{ of } 640 - 12.5\% \text{ of } 352 + 25\% \text{ of } 640 = ? - 48\% \text{ of } 432$$

$$230.4 - 44 + 160 = ? - 207.36$$

$$346.4 = ? - 207.36$$

$$? = 346.4 + 207.36 = 553.76$$

Sol- 41. 4

Sol- 42. 1

Sol- 43. 5

Sol- 44. 4

Sol- 45. 4

Sol- 46. 1

Sol- 47. 4

Sol- 48. 4

Sol- 49. 2

Sol- 50. 2